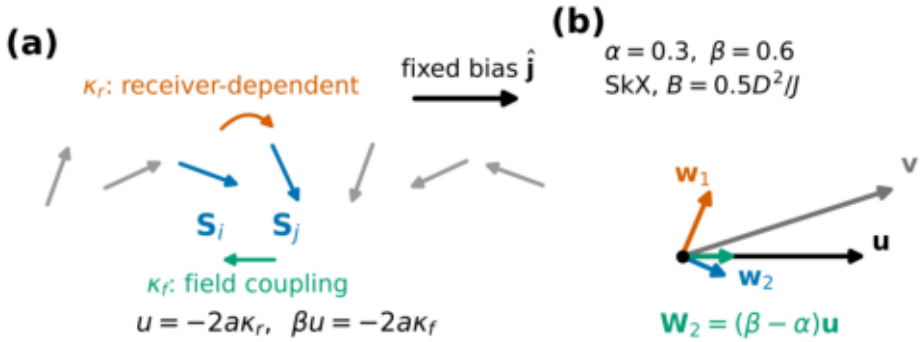


Fig. 1

A current is a fixed nonreciprocal bond bias, and in the co-moving frame it enters only through two vectors of opposite time-reversal parity. (a) On a bond i → j the adiabatic term is a receiver-dependent coupling κ_r and the β term a field coupling κ_f, with u = −2aκ_r and βu = −2aκ_f. (b) Slot vectors of the driven skyrmion lattice (B = 0.5D²/J, α = 0.3, β = 0.6) from the exact Thiele projection. $w_1 = v - u$ (T-odd) and $w_2 = \beta u - \alpha v$ (T-even) are perpendicular, and the Landau–Lifshitz drive $W_2 = w_2 + \alpha w_1 = (\beta - \alpha)u$ points along u whatever the Hall angle of v.

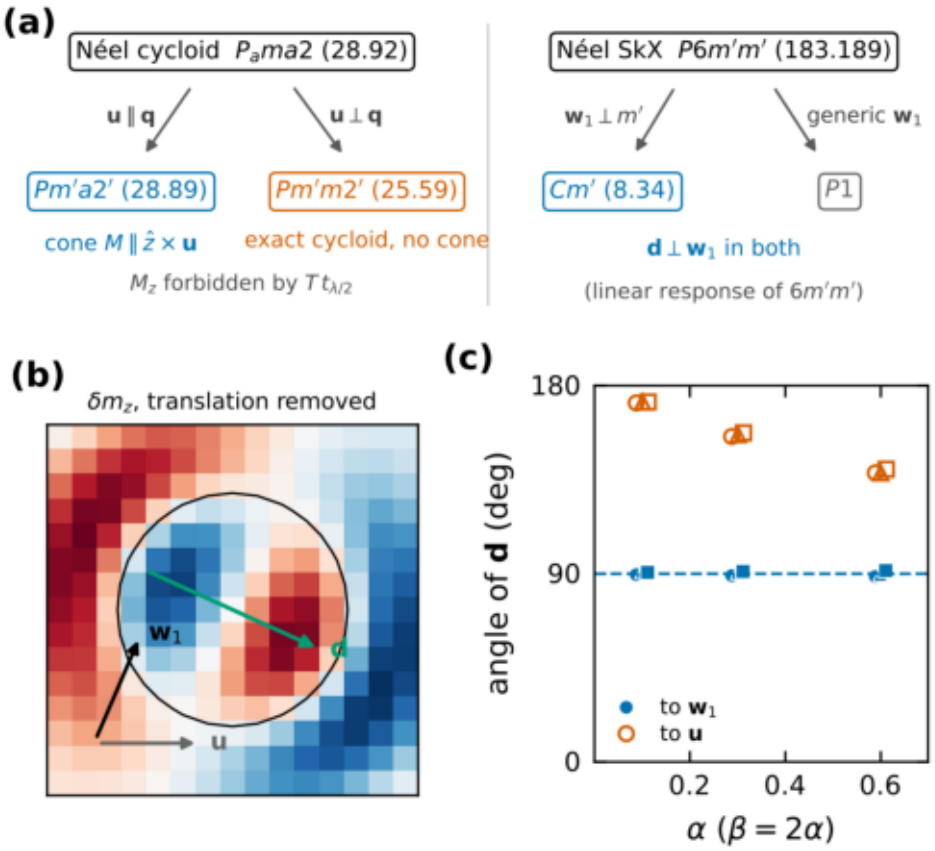


Fig. 2

The residual magnetic group of the driven pattern fixes its first-order response, and the vector that matters is w₁, not u. (a) Group–subgroup steps with BNS numbers (spglib on minimal commensurate cells). Cycloid: u ∥ q leaves Pm′a2′ and allows a uniform cone M ∥ ẑ×u; u ⊥ q leaves Pm′m2′ and the cycloid stays exact; the out-of-plane cone is forbidden by the half-period antitranslation T t_{λ/2}. Skyrmion lattice: linear response of 6m′m′ gives a dipolar distortion d = a ẑ×w₁ + b w₂, perpendicular to w₁. (b) Change of m_z of a driven Néel skyrmion (α = 0.3, u = 0.02 along x) after the uniform translation is removed. Black contour: m_z = 0 of the static skyrmion. Green arrow: first moment d inside r < 5. (c) Angle of d measured from w₁ (filled) and from u (open) for α = 0.1, 0.3, 0.6 and current directions 0°, 30°, 90° (circle, triangle, square). d stays perpendicular to w₁ within 1.5°; its angle to u follows the Hall rotation.

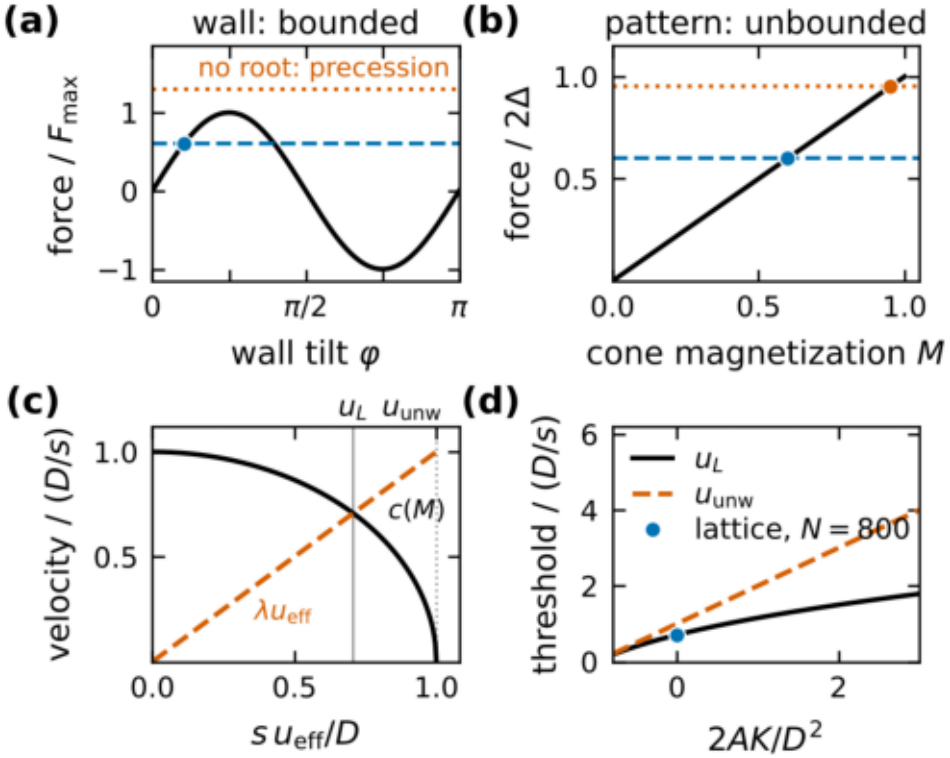


Fig. 3

A pattern has no precessional Walker breakdown; its static branch ends in a Landau instability before it can unwind. (a) The conjugate variable of a domain wall, the tilt φ, has a bounded restoring force. A drive above F_max (dotted) has no static root and the wall precesses. (b) The conjugate variable of a pattern, the cone M, has a quadratic energy, so a static root exists up to |M| = 1. (c) Helix with exchange and DMI: the compression velocity on the driven cone, c = (D/s)√(1 − M²) with M = su_eff/D, meets the Doppler line λu_eff (λ = 1) at u_L = D/(√2 s), below the unwinding point u_unw = D/s. (d) With easy-axis anisotropy K, u_L = Λ/√(1 + κ²) stays below u_unw = Λ/κ for every K (Λ = √(D² + 2AK)/s, κ = D/√(D² + 2AK)). Point: exact Bloch threshold of an N = 800 chain.

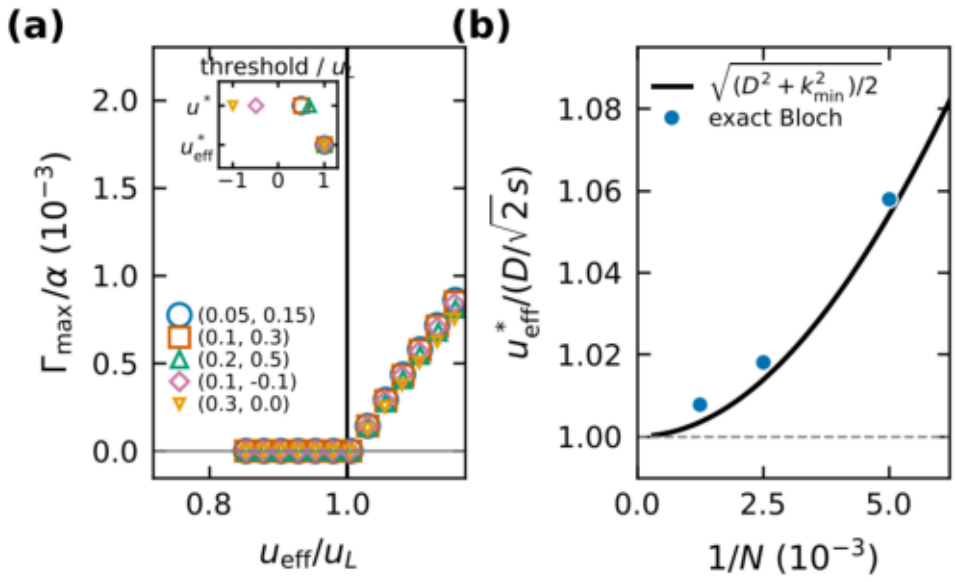


Fig. 4

The driven helix breaks at u_L = D/(√2 s) for every (α, β): the threshold depends on β only through u_eff = (β/α − 1)u. (a) Largest Bloch growth rate divided by α against u_eff for five (α, β), including β = 0 and β < 0 (N = 200; u_L includes the finite-chain correction √((D² + k_min²)/2)). Inset: the same thresholds in the bare current u (spread over a factor of two and both signs) and in u_eff (equal to 0.1%). (b) Finite chains: exact thresholds (points) against √((D² + k_min²)/2) (line), converging to D/√2 (dashed).

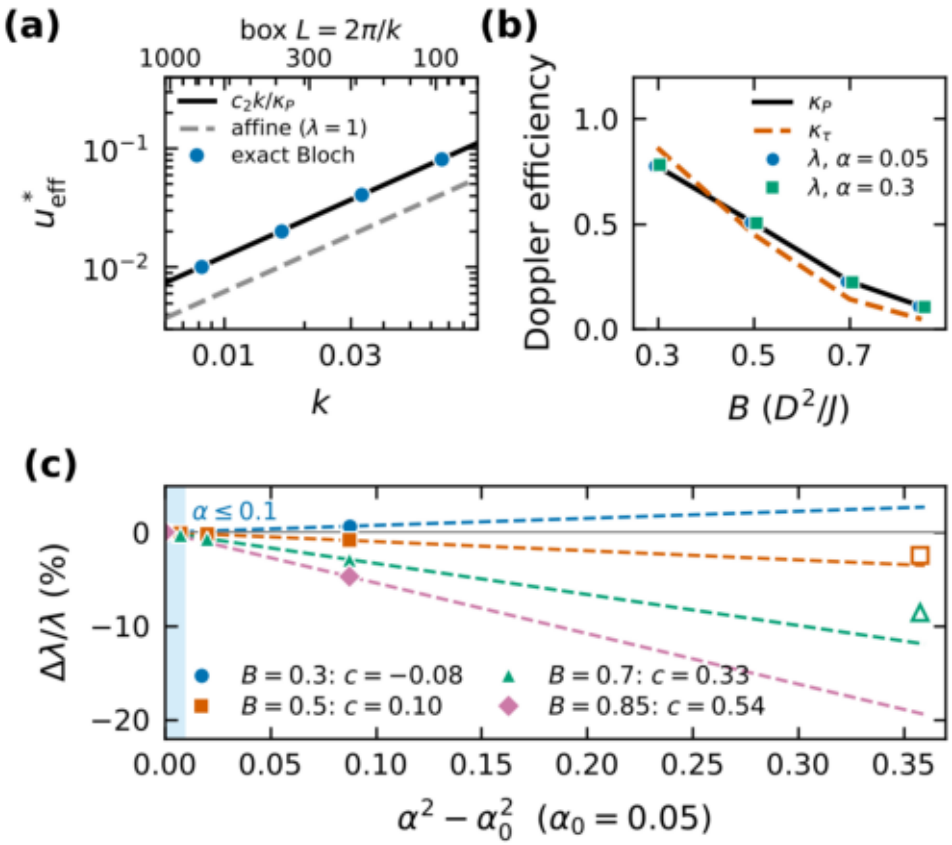


Fig. 5

The skyrmion-lattice threshold is u_eff* = c₂k/λ with λ = κ_P, the dissipative overlap; damping enters only at order α². (a) Exact Bloch thresholds (B = 0.5D²/J, α = 0.3, β = 0.6) at k = 2π/(19n), n = 40, 20, 10, 5, against c₂k/κ_P with c₂ and κ_P from the static lattice (line; agreement within 0.4–0.9%) and the affine estimate λ = 1 (dashed). Top axis: box length L = 2π/k, so the threshold falls as 1/L. (b) λ from exact thresholds at α = 0.05 and 0.3 lies on κ_P (solid), not on κ_τ (dashed), at four fields. (c) Relative change of λ from α₀ = 0.05, Δλ/λ = −c(α² − α₀²), fitted for α ≤ 0.3 (filled, dashed lines). Open symbols: α = 0.6, near critical damping. Shaded band: α ≤ 0.1, where λ = κ_P within 0.6%.

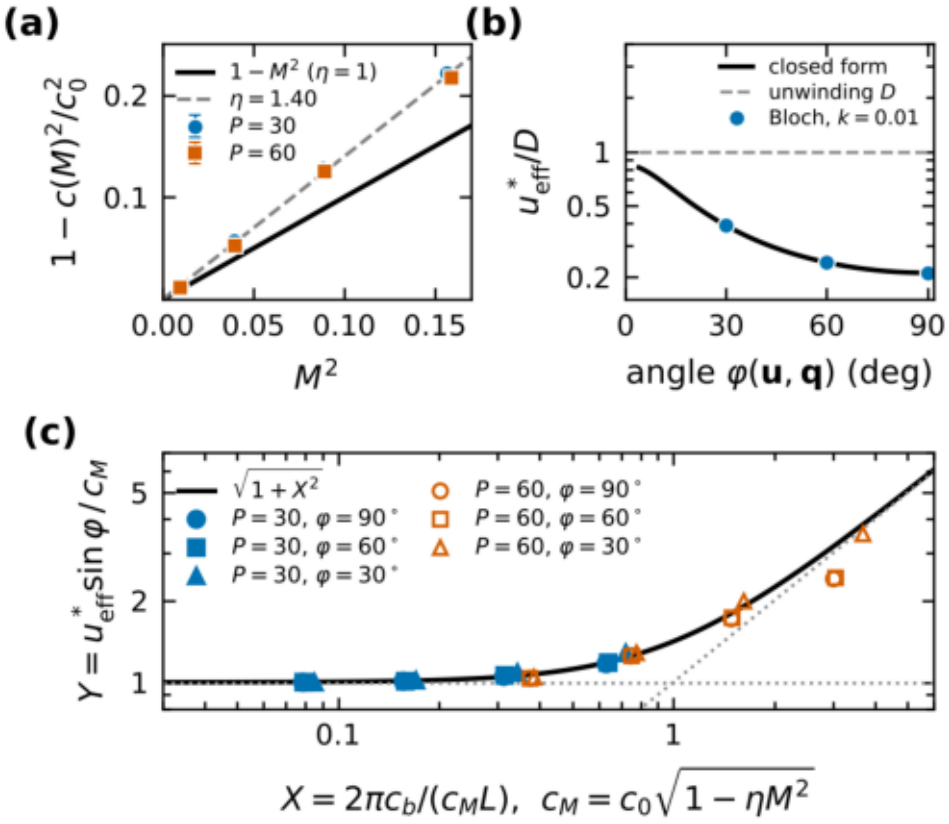


Fig. 6

Stripes have a quadratic Goldstone mode: a current component along q only scales the bending velocity, and every finite-box threshold collapses onto u_eff* sin φ = √(c_M² + (2πc_b/L)²). (a) k-resolved bending velocity on converged driven states (u ∥ q, residual < 10^{-12}; error bars: spread over k = 0.0025–0.04 q). $1 - c(M)^2/c_0^2 = \eta M^2$ with η ≈ 1.4, the same for every k and for P = 30 and 60, steeper than the exchange value η = 1 (solid). (b) Threshold against the current angle (P = 30): exact Bloch thresholds at k = 0.01 (points) and the k → 0 closed form $u^* = c_0/\sqrt{(\sin^2\varphi + \eta(c_0/D)^2\cos^2\varphi)}$ (line). (c) Dynamical thresholds in sheared boxes of length L = 64–512 (request 6) for P = 30 (filled) and 60 (open) and φ = 90°, 60°, 30° collapse onto $Y = \sqrt{(1 + X^2)}$, with $c_M = c_0\sqrt{(1 - \eta M^2)}$. The P = 60, L = 64 points leave the curve where the box wavenumber approaches the stripe wavenumber.

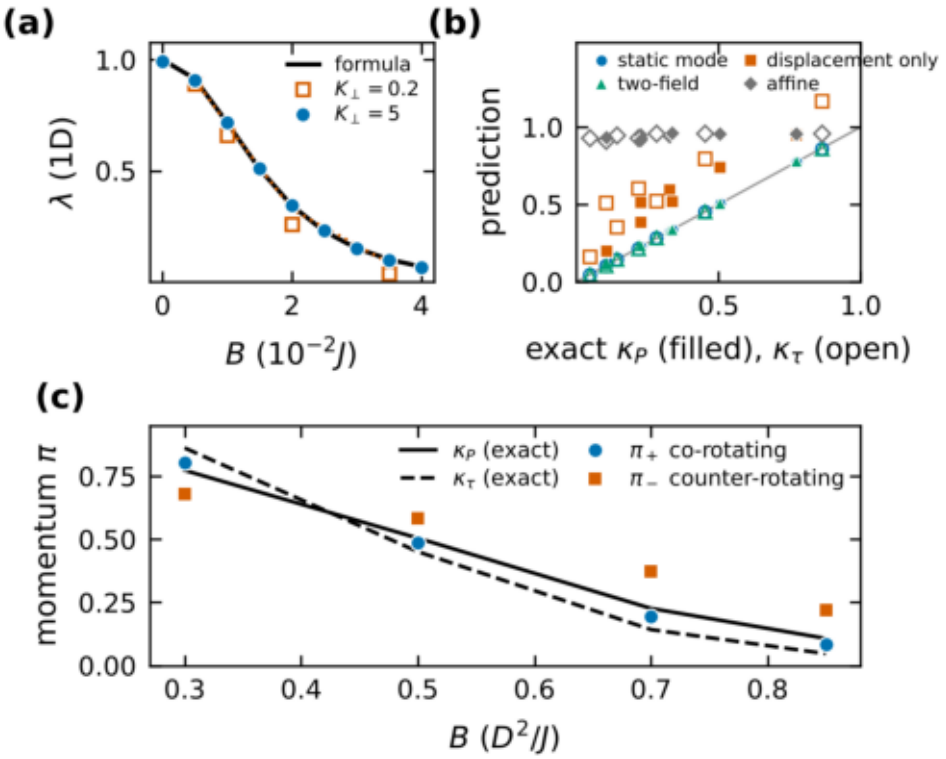


Fig. 7

λ is a static property of the pattern: one minus the counterflow momentum carried by its internal relaxation. (a) 1D chiral soliton lattice (D = 0.3, period 40): exact Doppler overlap against the homogenization formula $\langle 1/M\rangle/(\langle\psi_0^{-2}\rangle\langle\psi_0^2/M\rangle)$ with a slaved conjugate field ($K_\perp = 5$). With $K_\perp = 0.2$ the conjugate field is not slaved and the formula (dotted) overestimates λ. (b) Seven skyrmion lattices: predictions against the exact dynamic overlaps κ_P (filled) and κ_τ (open). The static relaxed mode and the two-field Ritz theory (displacement plus canting, no fitted parameter) fall on the diagonal; displacement-only theory and the affine mode do not. (c) Circular decomposition of the Bloch mode (19×33): the counterflow cancels mainly the co-rotating momentum π_+; the counter-rotating π_- keeps more, so $\kappa_P > \kappa_\tau$ once $\pi_- > \pi_+$.

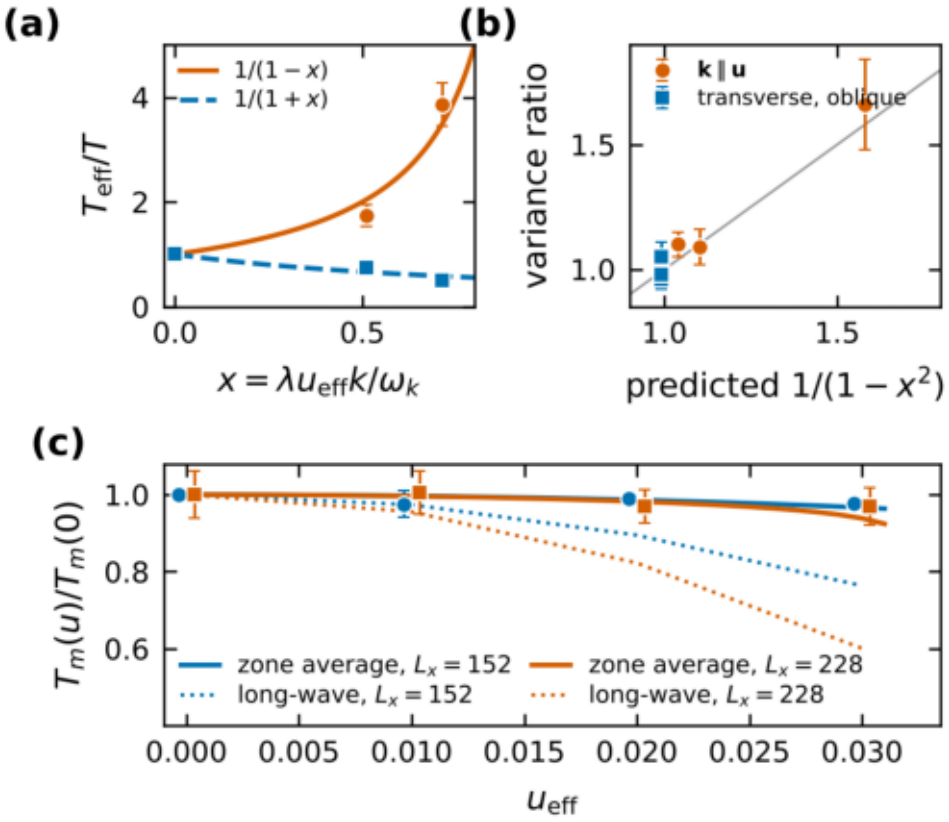


Fig. 8

Below threshold the current puts the lattice phonons in a Gibbs ensemble moving at λu_eff, and the melting point moves only by the zone average of that heating. (a) Effective temperatures of the two branches at $k_1 = 2\pi/76$ (stochastic LLG, 76×33, k_BT = 0.01) against $1/(1 \mp x)$. (b) Variance ratio of displacement modes in a 152×66 box at u_eff = 0.03 against $1/(1 - x^2)$. Modes along the current heat; transverse and oblique modes stay at the bath temperature. (c) Melting temperature from heating ladders (256–264 skyrmions, three seeds) against the zone average $1/\langle 1/(1 - x^2)\rangle_{BZ}$ (solid) and the rejected long-wave criterion (four longest modes, dotted), for two box shapes.

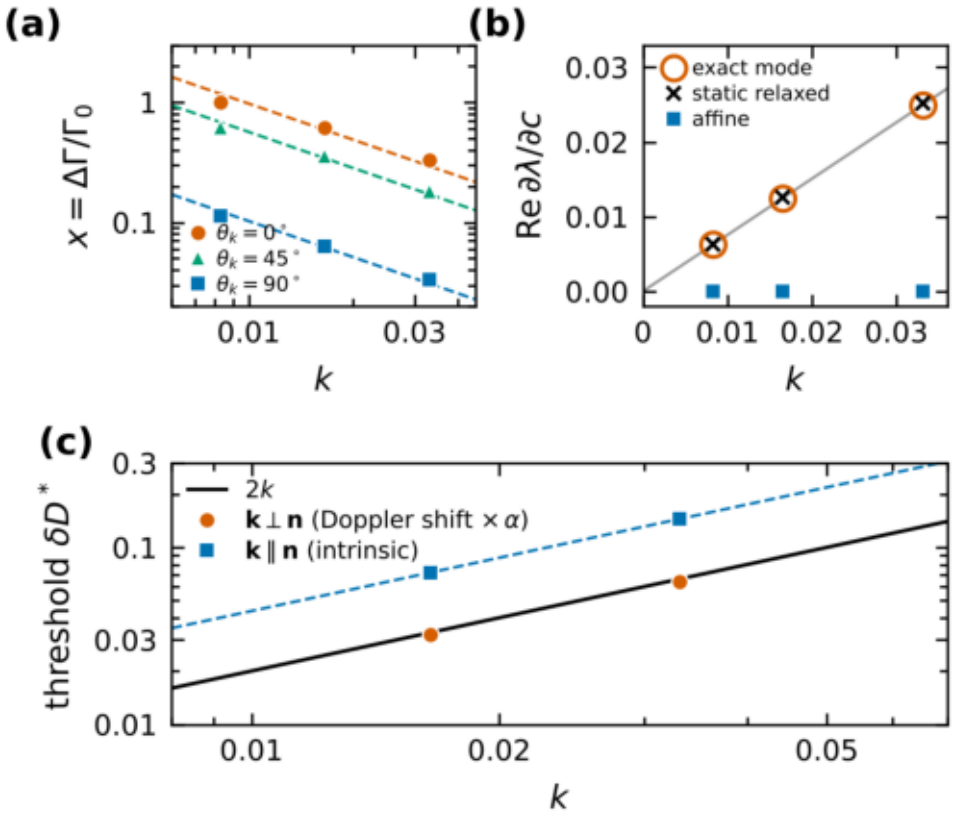


Fig. 9

Exchange-even couplings pump the lattice phonons with zero threshold through the static relaxed mode, the dual of the Zhang–Li Doppler term. (a) Damping-like SOT (c = 0.004, n = ŷ, α = 0.3): pumping ratio x = ΔΓ/Γ₀ against k for three directions of k. x·k is nearly constant (dashed: 1/k), so the threshold vanishes in an infinite system. (b) First-order generator Re ∂λ/∂c on the exact mode, the static relaxed mode and the affine mode (α = 0). The relaxed mode carries all the pumping; the affine mode carries none. (c) Second-order nonreciprocal stiffness of δD (n = ŷ, α = 0.3): threshold δD* = Γ₀/(half-split per δD) for k ⊥ n (Doppler shift converted by damping) and k ∥ n (intrinsic splitting). Both are ∝ k.